

CSE 321 STUDY QUESTION EXAMPLES

For each of the problems below, when you are asked to design a dynamic programming algorithm, you should answer the following questions.

- i. What is your decision sequence?
- ii. What are the alternatives at each decision step?
- iii. Show that principle of optimality is satisfied.
- iv. Write the recurrence relation based on principle of optimality.
- v. Describe the table that is necessary for dynamic programming algorithm. Describe how to fill the table.
- vi. Write down the pseudo-code of the dynamic programming algorithm.
- vii. Analyze the running time efficiency of your algorithm.
- viii. Write down the pseudo-code of an algorithm to find the actual solution.
- ix. Analyze the running time efficiency of your algorithm.

Question 1) You want to plant trees on your field, but the closest water source is D meters (e.g. 1350 meters) away from your field. There are a few of small basins (irrigation pools) between your field and the source but they are completely dried. There are L meters (e.g. 200 meters) length hoses in the basins, and they can be useful for transporting the water between the basins but the distances between them are shorter or longer than L meters (e.g. 200 meters). The guy in the hardware store will help to fit hoses between basins but he asks P liras (e.g. 2 TL) for each meter hose that he fixed (either to make the hose longer or shorter to fit it between basins). You want to minimize the money that you must pay. Therefore, design a dynamic programming algorithm that determines which basins must be used to pay lowest price to the store. For example, basins = [190, 220, 410, 580, 640, 770, 950, 1100] are distances of basins from the water source.

Question 2) You need to rent a cloud service for your new project. There are two companies on the market, and they offer different charges for each month (i.e., for each company, you are given an array of N elements representing the charges for following N months). As a wise person, you want to use the cheaper company for each month since both companies offer the same service. On the other hand, both companies ask for a fixed fee T to transfer your data from

the old service to their servers. Therefore, you need to optimize your monthly plan by considering the transfer costs. Design a dynamic programming algorithm that calculates the optimal plan (the plan with lowest cost).

Question 3) A palindrome is a word that reads the same backward and forward. The longest palindromic subsequence problem is the problem of finding the longest subsequence of a string that is also a palindrome. Propose a dynamic programming algorithm for the longest palindromic subsequence problem.

Question 4)

The following questions should be answered for each problem. Consider that there are additional questions about each problem under its definition.

- (a) Specify the decision steps that you use in your algorithm design.**
- (b) Specify alternatives at each decision step.**
- (c) What is the greedy choice that you use while selecting one alternative in your algorithm?**
- (d) Why may this greedy choice lead to an optimal solution?**
- (e) Design a greedy algorithm for the problem (You should give a pseudocode).**
- (f) Analysis time and space complexity of your algorithm.**
- (g) Does your algorithm always provide an optimal?**

4.1. John is a manager of a factory. He wants to pay some premium as an encouragement award to his employees with respect to their performances in the last month. All the employees sit in a line and each of them has a rating score according to his or her performance in the last month. John wants to give at least \$100 to each employee. If two employees sit next to each other, then the one with the higher rating must get more money. John wants to minimize the total money he must pay.

Assume that the number of employees n and the array of integers which represents the ratings of each employee arr are given. For example, he has 6 employees ($n = 6$) and his employees' ratings are $arr = [4, 6, 4, 5, 6, 2]$. As a solution, he gives the money in the following distribution as dollars: $[100, 200, 100, 200, 300, 100]$. Hence, he must pay \$1000 as the minimum amount. (Note that when two employees have equal rating, they are allowed to have different amount of money and the positions of the rating values in the arr cannot be changed. Amount of money that is given to each employee must be multiples of 100.)

(h) Apply the following instances to your algorithm (Show your work step by step):

- $n = 10$
- $arr = [2, 4, 2, 6, 1, 7, 8, 9, 2, 1]$

(i) What is the distribution of money that John must pay for the given instances in (h) when you apply your algorithm?

(j) What is the total amount of money that John must pay for the given instances in (h)?

4.2. Fast Track is a transportation company and it has trucks which are able to move items from a city to another city. Assume that:

- There are n number of items to transfer within the trucks. The items have known weight values which are represented as $w_1, w_2, \dots, w_n$. (The items are one-dimensional and no need to rotate them.)
 - All the trucks have a equal capacity value M .
 - The total weight value of the items in a truck W cannot exceed the capacity of the truck M ($W \leq M$).
- The company aims to use **the minimum number of the trucks** while moving n items from a city to another city.

(h) Apply your algorithm to the following instances (Show your work step by step):

- There are 9 items and their weight values are $w_1 = 5, w_2 = 7, w_3 = 3, w_4 = 5, w_5 = 12, w_6 = 11, w_7 = 10, w_8 = 11, w_9 = 9$.
- The capacity value of each truck: $M = 14$.

(i) How many trucks are used to fit all the items in (h)?

(j) Does your algorithm give the optimal solution for the given instances in (h)?